Connor J. Haynes

Department of Mathematics, University of Western Ontario, London, ON, Canada.
Email: chayne8@uwo.ca **Website:** https://connorjhaynes.github.io/

Mathematical interests

Broadly, combinatorics, algebraic geometry, polyhedral geometry, algebraic combinatorics, elimination theory, and the application of these tools to problems in the sciences.

Specifically, matroid theory, Baker-Bowler theory, tropical geometry, combinatorics of generic sparse systems, Combinatorial Hodge Theory, geometric tomography, and the application of these tools to problems in the sciences.

Education

MSc in Pure Mathematics, University of Western Ontario — Aug. 2026 – present

Advisor: Taylor Brysiewicz
Thesis title (in progress): *Algebraic matroids of generic sparse systems*

BSc in Mathematics, Georgia Institute of Technology — May 2022 – May 2025

Graduated with Highest Honors

Workshops

University of Notre Dame
Center for Mathematics at Notre Dame: Thematic Program in Algebraic Combinatorics and Applications — June - July 2026

Georgia Institute of Technology
Future of Macaulay2 Workshop — May 2026

University of Notre Dame
Center for Mathematics at Notre Dame: Thematic Program in Discrete Groups in Topology and Algebraic Geometry — June - July 2025

Louisiana State University
RTG Summer Workshop: Matroids and Hyperplane Arrangements — May 2025

Teaching

Graduate Teaching Assistant, University of Western Ontario
MATH 1228 - Methods of Finite Mathematics — 2026

Undergraduate Teaching Assistant, Georgia Institute of Technology
MATH 1554 - Linear Algebra — 2024
MATH 1552 - Integral Calculus — 2025

Recipient of inaugural **Outstanding Undergraduate TA Lab Tutor** award

Student Assistant, Georgia Institute of Technology
MATH 3012 - Applied Combinatorics — 2025

Service

Mentor, University of Western Ontario
Undergraduate Directed Reading Program — 2026 - present

Programming languages

Computational algebraic geometry, polyhedral geometry, and combinatorics in `Julia`, `Macaulay2`, and `Sage`.

Modelling in `Python`.